

Отчёт по лабораторной работе № 6

Название работы: Интерактивный rebase, cherry-pick, теги, reflog

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1. Выполнение шагов лабораторной работы

Отметьте галочками выполненные шаги. В столбце «Команды / вывод» напишите выполненные команды и полученный вывод (поле автоматически растягивается по содержимому).

Шаг	Описание	Вып.	Команды / вывод
1	Настроить редактор Git на nano (git config --global core.editor 'nano')	☒	Поставил Notepad
2	Создать репозиторий и сделать 5 коммитов (Step 1..5)	☒	PS D:\Projects\URFU\GitLearn\lab6> 1..5 ForEach-Object { \$i = \$_; "Commit \$i" Out-File -FilePath "f\$i.txt" -Encoding utf8; git add .; git commit -m "Step \$i" } [master (root-commit) 22d3dc1] Step 1 1 file changed, 1 insertion(+) create mode 100644 f1.txt [master e53bd0e] Step 2 1 file changed, 1 insertion(+) create mode 100644 f2.txt [master cf4a022] Step 3 1 file changed, 1 insertion(+) create mode 100644 f3.txt [master 01a3b0d] Step 4 1 file changed, 1 insertion(+) create mode 100644 f4.txt [master 783ff1a] Step 5 1 file changed, 1 insertion(+) create mode 100644 f5.txt PS D:\Projects\URFU\GitLearn\lab6>

Шаг	Описание	Вып.	Команды / вывод
3	Создать ветку feature-rebase, добавить три коммита (Feature A, Feature B, Fix typo)	☒	<pre> PS D:\Projects\URFU\GitLearn\lab6> git checkout -b feature-rebase Switched to a new branch 'feature-rebase' PS D:\Projects\URFU\GitLearn\lab6> echo "Feature A" > fa.txt; git add .; git commit -m "Feature A" [feature-rebase 60ae8eb] Feature A 1 file changed, 0 insertions(+), 0 deletions(-) create mode 100644 fa.txt PS D:\Projects\URFU\GitLearn\lab6> echo "Feature B" > fb.txt; git add .; git commit -m "Feature B" [feature-rebase 99f7f3f] Feature B 1 file changed, 0 insertions(+), 0 deletions(-) create mode 100644 fb.txt PS D:\Projects\URFU\GitLearn\lab6> echo "Fix" >> f3.txt; git commit -am "Fix typo" [feature-rebase 6844b48] Fix typo 1 file changed, 0 insertions(+), 0 deletions(-) PS D:\Projects\URFU\GitLearn\lab6> git log --oneline 6844b48 (HEAD -> feature-rebase) Fix typo 99f7f3f Feature B 60ae8eb Feature A 783ff1a (master) Step 5 01a3b0d Step 4 cf4a022 Step 3 e53bd0e Step 2 22d3dc1 Step 1 </pre>
4	Запустить git rebase -i HEAD~3 и изменить: squash для Feature B, reword для Fix typo	☒	<pre> PS D:\Projects\URFU\GitLearn\lab6> git rebase -i HEAD~3 hint: Waiting for your editor to close the file... unix2dos: converting file D:/Projects/URFU/GitLearn/lab6/.git/reb ase-merge/git-rebase-todo to DOS format... dos2unix: converting file </pre>

Шаг	Описание	Вып.	Команды / вывод
			D:/Projects/URFU/GitLearn/lab6/.git/rebase-merge/git-rebase-todo to Unix format... hint: Waiting for your editor to close the file... unix2dos: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to DOS format... dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to Unix format... [detached HEAD e53c61b] Feature A Date: Thu May 14 20:01:11 2026 +0500 2 files changed, 0 insertions(+), 0 deletions(-) create mode 100644 fa.txt create mode 100644 fb.txt hint: Waiting for your editor to close the file... unix2dos: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to DOS format... dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to Unix format... [detached HEAD 6282f51] Fix critical typo in f3.txt Date: Thu May 14 20:01:30 2026 +0500 1 file changed, 0 insertions(+), 0 deletions(-) Successfully rebased and updated refs/heads/feature-rebase. PS D:\Projects\URFU\GitLearn\lab6> git log --oneline 6282f51 (HEAD -> feature-rebase) Fix critical typo in f3.txt e53c61b Feature A 783ff1a (master) Step 5 01a3b0d Step 4 cf4a022 Step 3 e53bd0e Step 2 22d3dc1 Step 1 PS D:\Projects\URFU\GitLearn\lab6> git rebase -i HEAD~3 hint: Waiting for your editor to close the

Шаг	Описание	Вып.	Команды / вывод
			file... unix2dos: converting file D:/Projects/URFU/GitLearn/lab6/.git/rebase-merge/git-rebase-todo to DOS format... dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/rebase-merge/git-rebase-todo to Unix format... Successfully rebased and updated refs/heads/feature-rebase. PS D:\Projects\URFU\GitLearn\lab6>
5	Сохранить изменения в редакторе (nano: Ctrl+O, Enter, Ctrl+X)	☒	Done
6	При squash объединить сообщения коммитов, при reword написать 'Fix critical typo in f3.txt'	☒	PS D:\Projects\URFU\GitLearn\lab6> git rebase -i HEAD~3 hint: Waiting for your editor to close the file... unix2dos: converting file D:/Projects/URFU/GitLearn/lab6/.git/rebase-merge/git-rebase-todo to DOS format... dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/rebase-merge/git-rebase-todo to Unix format... hint: Waiting for your editor to close the file... unix2dos: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to DOS format... dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to Unix format... [detached HEAD e53c61b] Feature A Date: Thu May 14 20:01:11 2026 +0500 2 files changed, 0 insertions(+), 0 deletions(-) create mode 100644 fa.txt create mode 100644 fb.txt hint: Waiting for your editor to close the file... unix2dos: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to DOS format...

Шаг	Описание	Вып.	Команды / вывод
			dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to Unix format... [detached HEAD 6282f51] Fix critical typo in f3.txt Date: Thu May 14 20:01:30 2026 +0500 1 file changed, 0 insertions(+), 0 deletions(-) Successfully rebased and updated refs/heads/feature-rebase.
7	Проверить историю (git log --oneline) — коммитов стало меньше	☒	PS D:\Projects\URFU\GitLearn\lab6> git log --oneline 6282f51 (HEAD -> feature-rebase) Fix critical typo in f3.txt e53c61b Feature A 783ff1a (master) Step 5 01a3b0d Step 4 cf4a022 Step 3 e53bd0e Step 2 22d3dc1 Step 1
8	Переключиться в main и скопировать исправленный коммит через cherry-pick	☒	PS D:\Projects\URFU\GitLearn\lab6> git checkout master Switched to branch 'master' PS D:\Projects\URFU\GitLearn\lab6> git log --oneline feature-rebase 6282f51 (feature-rebase) Fix critical typo in f3.txt e53c61b Feature A 783ff1a (HEAD -> master) Step 5 01a3b0d Step 4 cf4a022 Step 3 e53bd0e Step 2 22d3dc1 Step 1 PS D:\Projects\URFU\GitLearn\lab6> git cherry-pick 6282f51 [master ea4ab73] Fix critical typo in f3.txt Date: Thu May 14 20:01:30 2026 +0500 1 file changed, 0 insertions(+), 0

Шаг	Описание	Вып.	Команды / вывод
			deletions(-) PS D:\Projects\URFU\GitLearn\lab6> git log --oneline ea4ab73 (HEAD -> master) Fix critical typo in f3.txt 783ff1a Step 5 01a3b0d Step 4 cf4a022 Step 3 e53bd0e Step 2 22d3dc1 Step 1
9	Создать лёгкий тег v1.0.0 и аннотированный тег v1.1.0	☒	PS D:\Projects\URFU\GitLearn\lab6> git tag v1.0.0 PS D:\Projects\URFU\GitLearn\lab6> git tag -a v1.1.0 -m "Аннотированный тег"
10	Переключиться по тегу v1.0.0 (detached HEAD), создать от него ветку hotfix-from-tag	☒	PS D:\Projects\URFU\GitLearn\lab6> git checkout v1.0.0 # detached HEAD Note: switching to 'v1.0.0'. You are in 'detached HEAD' state. You can look around, make experimental changes and commit them, and you can discard any commits you make in this state without impacting any branches by switching back to a branch. If you want to create a new branch to retain commits you create, you may do so (now or later) by using -c with the switch command. Example: git switch -c <new-branch-name> Or undo this operation with: git switch - Turn off this advice by setting config variable advice.detachedHead to false HEAD is now at ea4ab73 Fix critical

Шаг	Описание	Вып.	Команды / вывод
			typo in f3.txt PS D:\Projects\URFU\GitLearn\lab6> git checkout -b hotfix-from-tag # создать ветку от тега Switched to a new branch 'hotfix-from-tag'
11	Выполнить git reset --hard HEAD~3, затем восстановить потерянные коммиты через reflog	✓	PS D:\Projects\URFU\GitLearn\lab6> git reset --hard HEAD~3 HEAD is now at cf4a022 Step 3 PS D:\Projects\URFU\GitLearn\lab6> git log --oneline cf4a022 (HEAD -> master) Step 3 e53bd0e Step 2 22d3dc1 Step 1 PS D:\Projects\URFU\GitLearn\lab6> git reflog cf4a022 (HEAD -> master) HEAD@{0}: reset: moving to HEAD~3 ea4ab73 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{1}: checkout: moving from hotfix-from-tag to master ea4ab73 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{2}: checkout: moving from ea4ab7393f133ff4c360462b97c40e488 eaac297 to hotfix-from-tag ea4ab73 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{3}: checkout: moving from master to v1.0.0 ea4ab73 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{4}: cherry- pick: Fix critical typo in f3.txt 783ff1a HEAD@{5}: checkout: moving from feature-rebase to master 6282f51 (feature-rebase) HEAD@{6}: rebase (finish): returning to refs/heads/feature-rebase 6282f51 (feature-rebase) HEAD@{7}: rebase (start): checkout HEAD~3 6282f51 (feature-rebase) HEAD@{8}: rebase (finish): returning to refs/heads/feature-rebase 6282f51 (feature-rebase) HEAD@{9}: rebase (reword): Fix critical typo in

Шаг	Описание	Вып.	Команды / вывод
			f3.txt 21f4805 HEAD@{10}: rebase (reword): Fix typo e53c61b HEAD@{11}: rebase (squash): Feature A 60ae8eb HEAD@{12}: rebase (start): checkout HEAD~3 6844b48 HEAD@{13}: commit: Fix typo 99f7f3f HEAD@{14}: commit: Feature B 60ae8eb HEAD@{15}: commit: Feature A 783ff1a HEAD@{16}: checkout: moving from master to feature-rebase 783ff1a HEAD@{17}: commit: Step 5 01a3b0d HEAD@{18}: commit: Step 4 cf4a022 (HEAD -> master) HEAD@{19}: commit: Step 3 e53bd0e HEAD@{20}: commit: Step 2 22d3dc1 HEAD@{21}: commit (initial): Step 1 PS D:\Projects\URFU\GitLearn\lab6> git reset --hard HEAD@{1} error: unknown switch `e' PS D:\Projects\URFU\GitLearn\lab6> git reset --hard 'HEAD@{1}' HEAD is now at ea4ab73 Fix critical typo in f3.txt

2. Самостоятельное задание

Условие:

Создайте ветку messy, сделайте 4 коммита. С помощью git rebase -i HEAD~4 объедините первые два коммита и переименуйте третий. Перенесите один коммит из messy в main через cherry-pick. Создайте аннотированный тег exercise-v1.

Выполнение (ход действий и результат):

```
PS D:\Projects\URFU\GitLearn\lab6> 1..4| ForEach-Object { $i = $_; "Commit $i" | Out-File -
FilePath "f$i2.txt" -Encoding utf8; git add .; git commit -m "Step $i" }
[messy e11ac31] Step 1
1 file changed, 1 insertion(+)
create mode 100644 f.txt
[messy b98b600] Step 2
1 file changed, 1 insertion(+), 1 deletion(-)
```

[messy c2da5ba] Step 3

1 file changed, 1 insertion(+), 1 deletion(-)

[messy d751466] Step 4

1 file changed, 1 insertion(+), 1 deletion(-)

PS D:\Projects\URFU\GitLearn\lab6> git rebase -i HEAD~4

hint: Waiting for your editor to close the file... unix2dos: converting file

D:/Projects/URFU/GitLearn/lab6/.git/rebase-merge/git-rebase-todo to DOS format...

dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/rebase-merge/git-rebase-todo to Unix format...

hint: Waiting for your editor to close the file... unix2dos: converting file

D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to DOS format...

dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to Unix

format...[detached HEAD 8b1e3f4] Завершил первый этап работы

Date: Thu May 14 20:22:18 2026 +0500

1 file changed, 1 insertion(+)

create mode 100644 f.txt

hint: Waiting for your editor to close the file... unix2dos: converting file

D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to DOS format...

dos2unix: converting file D:/Projects/URFU/GitLearn/lab6/.git/COMMIT_EDITMSG to Unix

format...[detached HEAD 448e8d6] New Step 3

Date: Thu May 14 20:22:19 2026 +0500

1 file changed, 1 insertion(+), 1 deletion(-)

Successfully rebased and updated refs/heads/messy.

PS D:\Projects\URFU\GitLearn\lab6> git log --oneline

fb26cdb (HEAD -> messy) Step 4

448e8d6 New Step 3

8b1e3f4 Завершил первый этап работы

ea4ab73 (tag: v1.1.0, tag: v1.0.0, master, hotfix-from-tag) Fix critical typo in f3.txt

783ff1a Step 5

01a3b0d Step 4

cf4a022 Step 3

e53bd0e Step 2

22d3dc1 Step 1

PS D:\Projects\URFU\GitLearn\lab6> git checkout master

Switched to branch 'master'

PS D:\Projects\URFU\GitLearn\lab6> git cherry-pick 448e8d6

CONFLICT (modify/delete): f.txt deleted in HEAD and modified in 448e8d6 (New Step 3).

Version 448e8d6 (New Step 3) of f.txt left in tree.

error: could not apply 448e8d6... New Step 3

hint: After resolving the conflicts, mark them with

hint: "git add/rm <paths>", then run

hint: "git cherry-pick --continue".

hint: You can instead skip this commit with "git cherry-pick --skip".

hint: To abort and get back to the state before "git cherry-pick",

hint: run "git cherry-pick --abort".

hint: Disable this message with "git config set advice.mergeConflict false"

```
PS D:\Projects\URFU\GitLearn\lab6> git add .
PS D:\Projects\URFU\GitLearn\lab6> git status
On branch master
You are currently cherry-picking commit 448e8d6.
(all conflicts fixed: run "git cherry-pick --continue")
(use "git cherry-pick --skip" to skip this patch)
(use "git cherry-pick --abort" to cancel the cherry-pick operation)
```

Changes to be committed:

```
deleted:  f1.txt
deleted:  f2.txt
deleted:  f3.txt
deleted:  f4.txt
deleted:  f5.txt
```

```
PS D:\Projects\URFU\GitLearn\lab6> git commit -m "убрал всё чтоюы не мешалось"
```

```
[master 6ab6f81] убрал всё чтоюы не мешалось
```

```
Date: Thu May 14 20:22:19 2026 +0500
```

```
5 files changed, 4 deletions(-)
```

```
delete mode 100644 f1.txt
```

```
delete mode 100644 f2.txt
```

```
delete mode 100644 f3.txt
```

```
delete mode 100644 f4.txt
```

```
delete mode 100644 f5.txt
```

```
PS D:\Projects\URFU\GitLearn\lab6> git cherry-pick 448e8d6
```

```
CONFLICT (modify/delete): f.txt deleted in HEAD and modified in 448e8d6 (New Step 3).
```

```
Version 448e8d6 (New Step 3) of f.txt left in tree.
```

```
error: could not apply 448e8d6... New Step 3
```

```
hint: After resolving the conflicts, mark them with
```

```
hint: "git add/rm <paths>", then run
```

```
hint: "git cherry-pick --continue".
```

```
hint: You can instead skip this commit with "git cherry-pick --skip".
```

```
hint: To abort and get back to the state before "git cherry-pick",
```

```
hint: run "git cherry-pick --abort".
```

```
hint: Disable this message with "git config set advice.mergeConflict false"
```

```
PS D:\Projects\URFU\GitLearn\lab6> git checkout messy
```

```
f.txt: needs merge
```

```
error: you need to resolve your current index first
```

```
PS D:\Projects\URFU\GitLearn\lab6> git add .
```

```
PS D:\Projects\URFU\GitLearn\lab6> git commit -m "чë"
```

```
[master 7ee1cab] чë
```

```
Date: Thu May 14 20:22:19 2026 +0500
```

```
1 file changed, 1 insertion(+)
```

```
create mode 100644 f.txt
```

```
PS D:\Projects\URFU\GitLearn\lab6> git cherry-pick --continue
```

```
error: no cherry-pick or revert in progress
```

fatal: cherry-pick failed

```
PS D:\Projects\URFU\GitLearn\lab6> git cherry-pick 448e8d6
```

On branch master

You are currently cherry-picking commit 448e8d6.

(all conflicts fixed: run "git cherry-pick --continue")

(use "git cherry-pick --skip" to skip this patch)

(use "git cherry-pick --abort" to cancel the cherry-pick operation)

nothing to commit, working tree clean

The previous cherry-pick is now empty, possibly due to conflict resolution.

If you wish to commit it anyway, use:

```
git commit --allow-empty
```

Otherwise, please use 'git cherry-pick --skip'

```
PS D:\Projects\URFU\GitLearn\lab6> git branch
```

feature-rebase

hotfix-from-tag

* master

messy

```
PS D:\Projects\URFU\GitLearn\lab6> git tag -a exercise-v1 -m "Аннотированный тег"
```

```
PS D:\Projects\URFU\GitLearn\lab6>
```

3. Ответы на контрольные вопросы

1. **Вопрос 1:**

Ребейз переписывает историю коммитов и меняет их хэши. Вследствие этого получится неразбериха с другими пользователями, у которых сохранятся предыдущие хэши.

2. **Вопрос 2:**

Черри пик работает выборочно, а мердж берёт все коммиты и соединяет ветки

3. **Вопрос 3:**

git reflog , для последнего git reset --hard <хэш>

4. Дополнительные замечания (при необходимости)

```
git config --global core.editor "редактор"
```

Инструкция: Выберите номер лабораторной работы. Заполняйте поля — они автоматически растягиваются по высоте под текст. Для печати нажмите кнопку «Печать» — все рамки полей исчезнут, и отчёт будет выглядеть как обычный документ.